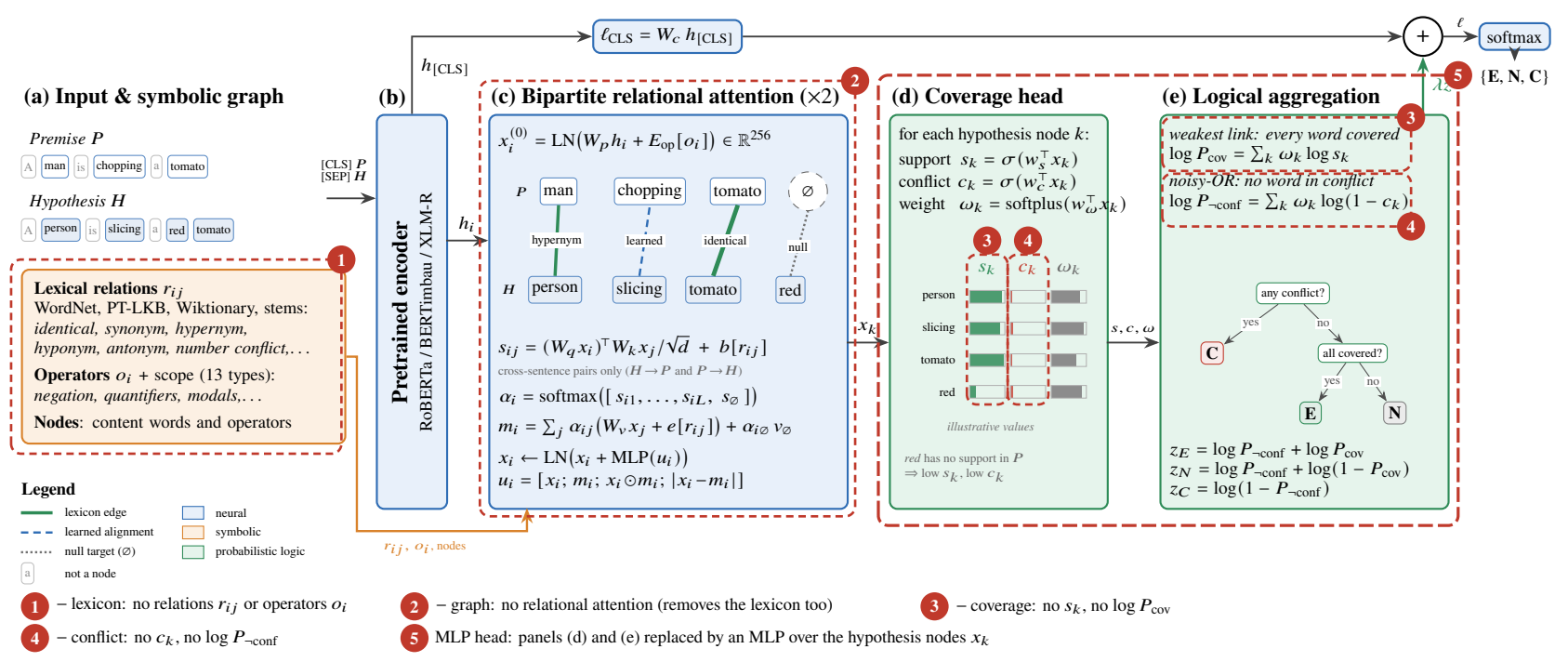

(a) Input & symbolic graph
Premise P
A man is chopping a tomato
Hypothesis H
A person is slicing a red tomato
1
Lexical relations r_{ij}
WordNet, PT-LKB, Wiktionary, stems:
identical, synonym, hypernym,
hyponym, antonym, number conflict,...
Operators o_i + scope (13 types):
negation, quantifiers, modals,...
Nodes: content words and operators
Legend
lexicon edge
learned alignment
null target (∅)
not a node
neural
symbolic
probabilistic logic
(b)
$h_{[CLS]}$
[CLS] P
[SEP] H
Pretrained encoder
RoBERTa / BERTimbau / XLM-R
h_i
r_{ij}, o_i, nodes
$\ell_{CLS} = W_c\, h_{[CLS]}$
(c) Bipartite relational attention (×2)
2
$x_i^{(0)} = \mathrm{LN}(W_p h_i + E_{op}[o_i]) \in \mathbb{R}^{256}$
P man chopping tomato ∅
hypernym learned identical null
H person slicing tomato red
$s_{ij} = (W_q x_i)^\top W_k x_j / \sqrt{d} + b[r_{ij}]$
cross-sentence pairs only ($H \to P$ and $P \to H$)
$\alpha_i = \mathrm{softmax}([\, s_{i1}, \ldots, s_{iL}, s_{i\varnothing} \,])$
$m_i = \sum_j \alpha_{ij} (W_v x_j + e[r_{ij}]) + \alpha_{i\varnothing} v_\varnothing$
$x_i \leftarrow \mathrm{LN}(x_i + \mathrm{MLP}(u_i))$
$u_i = [x_i;\, m_i;\, x_i \odot m_i;\, |x_i - m_i|]$
x_k
(d) Coverage head
for each hypothesis node k:
support $s_k = \sigma(w_s^\top x_k)$
conflict $c_k = \sigma(w_c^\top x_k)$
weight $\omega_k = \mathrm{softplus}(w_\omega^\top x_k)$
3
4
s_k c_k ω_k
person
slicing
tomato
red
illustrative values
red has no support in P
⇒ low s_k, low c_k
s, c, ω
(e) Logical aggregation
3
weakest link: every word covered
$\log P_{cov} = \sum_k \omega_k \log s_k$
noisy-OR: no word in conflict
$\log P_{\neg conf} = \sum_k \omega_k \log(1 - c_k)$
4
any conflict?
yes
no
C
all covered?
yes
no
E
N
$z_E = \log P_{\neg conf} + \log P_{cov}$
$z_N = \log P_{\neg conf} + \log(1 - P_{cov})$
$z_C = \log(1 - P_{\neg conf})$
ℓ
+
softmax
5
{E, N, C}
1 – lexicon: no relations r_{ij} or operators o_i
2 – graph: no relational attention (removes the lexicon too)
3 – coverage: no s_k, no $\log P_{cov}$
4 – conflict: no c_k, no $\log P_{\neg conf}$
5 MLP head: panels (d) and (e) replaced by an MLP over the hypothesis nodes x_k